

MIT 6.041SC Only Formula Notes
Lectures 1–25. Formula, one-line use, and symbols only.

Lecture 1: Probability Models and Axioms

Probability Axioms

Use: Use for checking whether a probability law is valid.

Formula:

$$\mathbb{P}(A) \geq 0, \quad \mathbb{P}(\Omega) = 1, \\ A_i \cap A_j = \emptyset \ (i \neq j) \implies \mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i).$$

Symbols:

- Ω is the sample space.
- A_i are events.

Complement and Inclusion–Exclusion

Use: Use when events overlap or the complement is easier.

Formula:

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A), \\ \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B). \\ \mathbb{P}(\text{at least one } A_i) = 1 - \mathbb{P}(\text{none of the } A_i).$$

Symbols:

- A^c means event A does not happen.
- $A \cup B$ means A or B .
- $A \cap B$ means both A and B .

Discrete Uniform Law

Use: Use when all outcomes are equally likely.

Formula:

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}.$$

Symbols:

- $|A|$ is the number of favorable outcomes.
- $|\Omega|$ is the total number of outcomes.

Lecture 2: Conditional Probability and Bayes

Conditional Probability

Use: Use when the problem says “given B ”.

Formula:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \quad \mathbb{P}(B) > 0.$$

Symbols:

- A is the event wanted.
- B is the condition/new universe.

Multiplication Rule

Use: Use for staged experiments or paths in a probability tree.

Formula:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A). \\ \mathbb{P}(A_1 \cap \cdots \cap A_n) = \mathbb{P}(A_1) \prod_{i=2}^n \mathbb{P}(A_i \mid A_1 \cap \cdots \cap A_{i-1}).$$

Symbols:

- A_i are events in sequence.

Total Probability

Use: Use when cases A_i form a partition.

Formula:

$$\mathbb{P}(B) = \sum_i \mathbb{P}(A_i)\mathbb{P}(B \mid A_i).$$

Symbols:

- A_i are disjoint and exhaustive cases.
- B is the event wanted.

Bayes’ Rule

Use: Use when the question reverses evidence and cause.

Formula:

$$\mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(A_i)\mathbb{P}(B \mid A_i)}{\sum_j \mathbb{P}(A_j)\mathbb{P}(B \mid A_j)}.$$

Symbols:

- A_i is one possible hidden case.
- B is observed evidence.
- $\mathbb{P}(A_i)$ is the prior probability.

Lecture 3: Independence

Event Independence

Use: Use when knowing one event gives no information about another.

Formula:

$$A, B \text{ independent} \iff \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B). \\ \mathbb{P}(A \mid B) = \mathbb{P}(A), \quad \mathbb{P}(B) > 0.$$

Symbols:

- A, B are events.

Conditional Independence

Use: Use when independence is checked inside a condition C .

Formula:

$$\mathbb{P}(A \cap B \mid C) = \mathbb{P}(A \mid C)\mathbb{P}(B \mid C).$$

Symbols:

- C is the condition.

Mutual Independence

Use: Use when more than two events must be independent together.

Formula:

$$\mathbb{P}\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} \mathbb{P}(A_i) \quad \text{for every subset } S.$$

Symbols:

- S is any subset of event indices.

Lecture 4: Counting

Basic Counting Rule

Use: Use when choices happen in stages.

Formula:

$$\# \text{ outcomes} = n_1 n_2 \cdots n_r.$$

Symbols:

- n_i is the number of choices at stage i .
- r is the number of stages.

Permutations

Use: Use when order matters.

Formula:

$$P(n, k) = \frac{n!}{(n - k)!}.$$

Symbols:

- n is total objects.
- k is number of ordered positions.

Combinations

Use: Use when order does not matter.

Formula:

$$\binom{n}{k} = \frac{n!}{k!(n - k)!}.$$

Symbols:

- n is total objects.
- k is number selected.

Lecture 5: Random Variables

PMF

Use: Use for a discrete random variable.

Formula:

$$p_X(x) = \mathbb{P}(X = x), \quad \sum_x p_X(x) = 1.$$

Symbols:

- X is a random variable.
- x is a possible value.

Expectation

Use: Use for long-run average value.

Formula:

$$\begin{aligned} \mathbb{E}[X] &= \sum_x x p_X(x). \\ \mathbb{E}[g(X)] &= \sum_x g(x) p_X(x). \end{aligned}$$

Symbols:

- $g(X)$ is a transformed random variable.

Variance

Use: Use for spread around the mean.

Formula:

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2. \\ \text{Var}(aX + b) &= a^2 \text{Var}(X). \end{aligned}$$

Symbols:

- a, b are constants.

Lecture 6: Conditional PMFs and Total Expectation

Conditional PMF

Use: Use for the distribution of X after event A .

Formula:

$$p_{X|A}(x) = \frac{\mathbb{P}(\{X = x\} \cap A)}{\mathbb{P}(A)}.$$

Symbols:

- A is the event conditioned on.

Conditional PMF Given Another RV

Use: Use when $Y = y$ is known.

Formula:

$$p_{X|Y}(x | y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

Symbols:

- $p_{X,Y}$ is the joint PMF.
- $p_Y(y)$ is the marginal PMF of Y .

Total Expectation

Use: Use when expectation is easy after conditioning on cases.

Formula:

$$\begin{aligned} \mathbb{E}[X] &= \sum_y \mathbb{E}[X | Y = y] p_Y(y). \\ \mathbb{E}[X] &= \mathbb{E}[\mathbb{E}[X | Y]]. \end{aligned}$$

Symbols:

- Y is the conditioning variable.

Lecture 7: Joint PMFs

Joint and Marginal PMFs

Use: Use when two discrete variables appear together.

Formula:

$$p_{X,Y}(x,y) = \mathbb{P}(X=x, Y=y).$$
$$p_X(x) = \sum_y p_{X,Y}(x,y), \quad p_Y(y) = \sum_x p_{X,Y}(x,y).$$

Symbols:

- $p_{X,Y}$ is the joint PMF.
- p_X, p_Y are marginal PMFs.

Function of Two RVs

Use: Use when $Z = g(X, Y)$.

Formula:

$$p_Z(z) = \sum_{(x,y): g(x,y)=z} p_{X,Y}(x,y).$$
$$\mathbb{E}[g(X, Y)] = \sum_x \sum_y g(x,y) p_{X,Y}(x,y).$$

Symbols:

- Z is the derived random variable.

Independent Random Variables

Use: Use when joint distribution factors.

Formula:

$$p_{X,Y}(x,y) = p_X(x)p_Y(y).$$
$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y], \quad \text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y).$$

Symbols:

- The variance formula needs independence.

Lecture 8: Continuous Random Variables

PDF and CDF

Use: Use for interval probabilities of continuous random variables.

Formula:

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx.$$
$$F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f_X(t) dt, \quad f_X(x) = F'_X(x).$$

Symbols:

- f_X is the PDF.
- F_X is the CDF.

Continuous Expectation and Variance

Use: Use continuous analogs of discrete expectation.

Formula:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx, \quad \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx.$$

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mathbb{E}[X])^2 f_X(x) dx.$$

Symbols:

- Integrals are over the support of X .

Uniform Distribution

Use: Use when all points in $[a, b]$ are equally dense.

Formula:

$$f_X(x) = \frac{1}{b-a}, \quad a \leq x \leq b.$$
$$\mathbb{E}[X] = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}.$$

Symbols:

- a, b are interval endpoints.

Normal Distribution

Use: Use when X is Gaussian or standardized for tables.

Formula:

$$X \sim N(\mu, \sigma^2), \quad Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$
$$\mathbb{P}(X \leq x) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$

Symbols:

- μ is the mean.
- σ^2 is the variance.
- Φ is the standard normal CDF.

Lecture 9: Joint PDFs and Continuous Conditioning

Joint PDF

Use: Use for probabilities over regions in the (x, y) plane.

Formula:

$$\mathbb{P}((X, Y) \in A) = \iint_A f_{X,Y}(x, y) dx dy.$$
$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx.$$

Symbols:

- A is a region in the plane.
- f_X, f_Y are marginal PDFs.

Conditional PDF

Use: Use when one continuous variable is observed.

Formula:

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$
$$\mathbb{E}[X | Y = y] = \int x f_{X|Y}(x | y) dx.$$

Symbols:

- $f_Y(y) > 0$ is required.

Continuous Independence

Use: Use when joint PDF factors into marginals.

Formula:

$$f_{X,Y}(x,y) = f_X(x)f_Y(y).$$

Symbols:

- The support must also factor consistently.

Lecture 10: Derived Distributions and Continuous Bayes

Monotone Transformation

Use: Use when $Y = g(X)$ and g is one-to-one.

Formula:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

Symbols:

- $g^{-1}(y)$ is the inverse value of x .

CDF Method

Use: Use when transformation is not one-to-one.

Formula:

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(g(X) \leq y), \quad f_Y(y) = \frac{d}{dy} F_Y(y).$$

Symbols:

- $Y = g(X)$ is the transformed variable.

Continuous Bayes

Use: Use when evidence is continuous.

Formula:

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x)f_X(x)}{\int f_{Y|X}(y|u)f_X(u)du}.$$

$$p_{X|Y}(x|y) = \frac{p_X(x)f_{Y|X}(y|x)}{\sum_u p_X(u)f_{Y|X}(y|u)}.$$

Symbols:

- First formula is continuous hidden X .
- Second formula is discrete hidden X .

Lecture 11: Convolution, Covariance, Correlation

Convolution

Use: Use for sum of independent random variables.

Formula:

$$W = X + Y, \quad X \perp Y.$$

$$p_W(w) = \sum_x p_X(x)p_Y(w-x).$$

$$f_W(w) = \int_{-\infty}^{\infty} f_X(x)f_Y(w-x)dx.$$

Symbols:

- $w - x$ is the value needed from Y .

Mixed Discrete–Continuous Sum

Use: Use when X is discrete and Y is continuous.

Formula:

$$f_W(w) = \sum_x p_X(x)f_Y(w-x).$$

Symbols:

- $W = X + Y$.

Covariance and Correlation

Use: Use for linear co-movement between variables.

Formula:

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y).$$

Symbols:

- $\sigma_X = \sqrt{\text{Var}(X)}$.

Lecture 12: Conditional Expectation and Total Variance

Conditional Expectation

Use: Use for the mean after observing $Y = y$.

Formula:

$$\mathbb{E}[X | Y = y] = \sum_x x p_{X|Y}(x | y).$$

$$\mathbb{E}[X | Y = y] = \int x f_{X|Y}(x | y) dx.$$

Symbols:

- Use the sum for discrete X .
- Use the integral for continuous X .

Total Variance

Use: Use when variance splits into within-case and between-case parts.

Formula:

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y]).$$

Symbols:

- $\mathbb{E}[\text{Var}(X | Y)]$ is average within-case variance.
- $\text{Var}(\mathbb{E}[X | Y])$ is variance of case means.

Lecture 13: Bernoulli Process

Bernoulli and Binomial

Use: Use Binomial when the total number of trials is fixed.

Formula:

$$X_i \sim \text{Bernoulli}(p), \quad \mathbb{P}(X_i = 1) = p.$$

$$S_n = \sum_{i=1}^n X_i \sim \text{Binomial}(n, p),$$

$$\mathbb{P}(S_n = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

$$\mathbb{E}[S_n] = np, \quad \text{Var}(S_n) = np(1-p).$$

Symbols:

- n is fixed number of trials.
- k is number of successes.
- p is success probability per trial.

First Success

Use: Use Geometric for the trial number of the first success.

Formula:

$$T_1 \sim \text{Geometric}(p), \quad \mathbb{P}(T_1 = t) = (1-p)^{t-1} p.$$

$$\mathbb{E}[T_1] = \frac{1}{p}, \quad \text{Var}(T_1) = \frac{1-p}{p^2}.$$

Symbols:

- T_1 counts the success trial.

Kth Success

Use: Use Negative Binomial/Pascal for the trial of the k th success.

Formula:

$$Y_k \sim \text{NegativeBinomial}(k, p),$$

$$\mathbb{P}(Y_k = t) = \binom{t-1}{k-1} p^k (1-p)^{t-k}, \quad t \geq k.$$

$$\mathbb{E}[Y_k] = \frac{k}{p}, \quad \text{Var}(Y_k) = \frac{k(1-p)}{p^2}.$$

Symbols:

- Y_k is the trial of the k th success.
- The final trial t is a success.

Lecture 14: Poisson Process I

Poisson Counts

Use: Use for continuous-time arrivals with rate λ .

Formula:

$$N(t) \sim \text{Poisson}(\lambda t), \quad \mathbb{P}(N(t) = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}.$$

$$\mathbb{E}[N(t)] = \lambda t, \quad \text{Var}(N(t)) = \lambda t.$$

Symbols:

- $N(t)$ is number of arrivals in length t .
- λ is arrival rate per unit time.

Small Interval Approximation

Use: Use for local behavior of a Poisson process.

Formula:

$$\mathbb{P}(N(\delta) = 0) \approx 1 - \lambda\delta, \quad \mathbb{P}(N(\delta) = 1) \approx \lambda\delta, \quad \mathbb{P}(N(\delta) \geq 2) \approx 0.$$

Symbols:

- δ is a very small time length.

Independent Increments

Use: Use for counts in non-overlapping intervals.

Formula:

$$N(t_2) - N(t_1) \sim \text{Poisson}(\lambda(t_2 - t_1)).$$

$N(t_2) - N(t_1)$ is independent of disjoint interval counts.

Symbols:

- $t_2 - t_1$ is the interval length.

Lecture 15: Poisson Process II

First Arrival Time

Use: Use Exponential for waiting time to the first Poisson arrival.

Formula:

$$T_1 \sim \text{Exponential}(\lambda), \quad f_{T_1}(t) = \lambda e^{-\lambda t}, \quad t \geq 0.$$

$$\mathbb{P}(T_1 > t) = e^{-\lambda t}, \quad \mathbb{E}[T_1] = \frac{1}{\lambda}, \quad \text{Var}(T_1) = \frac{1}{\lambda^2}.$$

Symbols:

- T_1 is time until first arrival.

Kth Arrival Time

Use: Use Erlang for waiting time to the k th Poisson arrival.

Formula:

$$Y_k \sim \text{Erlang}(k, \lambda), \quad f_{Y_k}(y) = \frac{\lambda^k y^{k-1} e^{-\lambda y}}{(k-1)!}.$$

$$\mathbb{E}[Y_k] = \frac{k}{\lambda}, \quad \text{Var}(Y_k) = \frac{k}{\lambda^2}.$$

Symbols:

- Y_k is time of the k th arrival.

Arrival Time and Count Conversion

Use: Use when converting k th-arrival questions into count questions.

Formula:

$$\mathbb{P}(Y_k \leq t) = \mathbb{P}(N(t) \geq k), \quad \mathbb{P}(Y_k > t) = \mathbb{P}(N(t) \leq k-1).$$

Symbols:

- $N(t)$ is the count by time t .

Merging and Splitting Poisson Processes

Use: Use merging for combined streams and splitting for typed arrivals.

Formula:

$N_i(t) \sim \text{Poisson}(\lambda_i t)$ independent,

$$\sum_i N_i(t) \sim \text{Poisson}\left(\sum_i \lambda_i t\right).$$

$$\mathbb{P}(\text{next arrival from stream } i) = \frac{\lambda_i}{\sum_j \lambda_j}.$$

$N_i(t) \sim \text{Poisson}(\lambda p_i t)$ after splitting by type i .

Symbols:

- λ_i is rate of stream i .
- p_i is type probability after splitting.

Conditioning on Total Poisson Count

Use: Use Binomial-style counting once total arrivals are fixed.

Formula:

$$N_1(t) \mid N(t) = m \sim \text{Binomial}(m, p_1).$$

Symbols:

- m is the given total count.
- p_1 is probability of type 1.

Random Incidence

Use: Use when observing a renewal process at a random time.

Formula:

$$\mathbb{P}(\text{chosen interval has length } \ell_i) = \frac{\ell_i p_i}{\mathbb{E}[L]}.$$

$$\mathbb{E}[\text{forward wait} \mid L = \ell_i] = \frac{\ell_i}{2}.$$

Symbols:

- L is ordinary interarrival length.
- $p_i = \mathbb{P}(L = \ell_i)$.

Lecture 16: Markov Processes I

Markov Property

Use: Use when the next state depends only on the current state.

Formula:

$$\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1}, \dots, X_0) = p_{ij}.$$

$$p_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i).$$

Symbols:

- X_n is the state at time n .
- p_{ij} is one-step transition probability.

N-Step Transitions

Use: Use for probability of being in state j after n steps.

Formula:

$$r_{ij}(n) = \mathbb{P}(X_n = j \mid X_0 = i) = (P^n)_{ij}.$$

$$r_{ij}(n) = \sum_k r_{ik}(n-1)p_{kj}.$$

Symbols:

- P is the transition matrix.
- $r_{ij}(n)$ is the n -step transition probability.

State Occupancy

Use: Use when the initial state is random.

Formula:

$$\pi^{(n)} = \pi^{(0)} P^n, \quad \mathbb{P}(X_n = j) = \sum_i \pi_i^{(0)} r_{ij}(n).$$

Symbols:

- $\pi^{(n)}$ is the row vector of state probabilities at time n .

Lecture 17: Markov Processes II

Steady-State Probabilities

Use: Use for long-run state probabilities in an aperiodic single recurrent class.

Formula:

$$\pi_j = \sum_i \pi_i p_{ij}, \quad \sum_j \pi_j = 1.$$
$$\pi = \pi P.$$

Symbols:

- π_j is steady-state probability of state j .

Long-Run Transition Frequency

Use: Use for fraction of transitions from i to j .

Formula:

$$\text{long-run frequency of } i \rightarrow j = \pi_i p_{ij}.$$

Symbols:

- π_i is long-run fraction of time in state i .

Birth–Death Chains

Use: Use for nearest-neighbor Markov chains.

Formula:

$$\pi_i p_i = \pi_{i+1} q_{i+1}.$$

$$\pi_i = \rho^i \pi_0, \quad \rho = \frac{p}{q} \quad \text{when } p_i = p, q_i = q.$$

$$\pi_0 = \frac{1 - \rho}{1 - \rho^{m+1}} \quad (\rho \neq 1, 0 \leq i \leq m).$$

$$\pi_i = (1 - \rho) \rho^i \quad (\rho < 1, i = 0, 1, \dots).$$

Symbols:

- p_i is transition probability from i to $i+1$.
- q_i is transition probability from i to $i-1$.
- ρ is the load ratio.

Lecture 18: Markov Processes III

Phone Blocking Model

Use: Use for B lines, Poisson arrivals, exponential call durations.

Formula:

$$\begin{aligned}\lambda\pi_{i-1} &= i\mu\pi_i, & i &= 1, \dots, B. \\ \pi_i &= \frac{(\lambda/\mu)^i}{i!}\pi_0, & \pi_0 &= \left(\sum_{i=0}^B \frac{(\lambda/\mu)^i}{i!}\right)^{-1}. \\ \mathbb{P}(\text{blocked}) &= \pi_B.\end{aligned}$$

Symbols:

- λ is call arrival rate.
- μ is service completion rate per active call.
- B is number of lines.

Absorption Probability

Use: Use for probability of eventually reaching an absorbing target.

Formula:

$$\begin{aligned}a_i &= \sum_j p_{ij}a_j & \text{for nonabsorbing states } i. \\ a_i &= 1 & \text{on target absorbing states,} \\ a_i &= 0 & \text{on other absorbing states.}\end{aligned}$$

Symbols:

- a_i is absorption probability starting from state i .

Expected Time to Absorption

Use: Use for expected number of transitions until absorption.

Formula:

$$\begin{aligned}t_i &= 1 + \sum_j p_{ij}t_j & \text{for nonabsorbing states } i, \\ t_i &= 0 & \text{on absorbing states.}\end{aligned}$$

Symbols:

- t_i is expected time to absorption from state i .

Mean First Passage and Recurrence

Use: Use for expected time to hit a state s .

Formula:

$$\begin{aligned}t_s &= 0, & t_i &= 1 + \sum_j p_{ij}t_j, & i &\neq s. \\ t_s^* &= 1 + \sum_j p_{sj}t_j.\end{aligned}$$

Symbols:

- t_i is mean first passage time from i to s .
- t_s^* is mean recurrence time of s .

Lecture 19: Chebyshev and WLLN

Chebyshev's Inequality

Use: Use for probability bounds using only mean and variance.

Formula:

$$\begin{aligned}\mathbb{P}(|X - \mu| \geq c) &\leq \frac{\sigma^2}{c^2}. \\ \mathbb{P}(|X - \mu| \geq k\sigma) &\leq \frac{1}{k^2}.\end{aligned}$$

Symbols:

- $\mu = \mathbb{E}[X]$.
- $\sigma^2 = \text{Var}(X)$.

Convergence in Probability

Use: Use when random variables concentrate near a number.

Formula:

$$Y_n \xrightarrow{p} a \iff \lim_{n \rightarrow \infty} \mathbb{P}(|Y_n - a| \geq \epsilon) = 0 \quad \forall \epsilon > 0.$$

Symbols:

- a is the deterministic limit.

Weak Law of Large Numbers

Use: Use for the sample mean of i.i.d. variables.

Formula:

$$\begin{aligned}M_n &= \frac{X_1 + \dots + X_n}{n}, & \mathbb{E}[M_n] &= \mu, & \text{Var}(M_n) &= \frac{\sigma^2}{n}. \\ \mathbb{P}(|M_n - \mu| \geq \epsilon) &\leq \frac{\sigma^2}{n\epsilon^2}, & M_n &\xrightarrow{p} \mu.\end{aligned}$$

Symbols:

- X_i are i.i.d. with mean μ and variance σ^2 .

Lecture 20: Central Limit Theorem

CLT Standardization

Use: Use for sums of many i.i.d. variables with finite variance.

Formula:

$$\begin{aligned}S_n &= X_1 + \dots + X_n, & Z_n &= \frac{S_n - n\mu}{\sigma\sqrt{n}}. \\ \mathbb{P}(Z_n \leq z) &\rightarrow \Phi(z). \\ S_n &\approx N(n\mu, n\sigma^2), & M_n &\approx N\left(\mu, \frac{\sigma^2}{n}\right).\end{aligned}$$

Symbols:

- Φ is the standard normal CDF.
- $M_n = S_n/n$.

Normal Approximation to Binomial

Use: Use for large n Binomial counts.

Formula:

$$S_n \sim \text{Binomial}(n, p), \quad \frac{S_n - np}{\sqrt{np(1-p)}} \approx N(0, 1).$$

$$\mathbb{P}(S_n \leq m) \approx \Phi\left(\frac{m + 0.5 - np}{\sqrt{np(1-p)}}\right).$$

Symbols:

- 0.5 is the continuity correction.

Poisson Approximation and Normal Approximation

Use: Use Poisson for rare Binomial events and Normal for large counts.

Formula:

$$\begin{aligned} \text{Binomial}(n, p) &\approx \text{Poisson}(\lambda), & \lambda = np \quad (n \text{ large, } p \text{ small}). \\ \text{Poisson}(\lambda) &\approx N(\lambda, \lambda) \quad (\lambda \text{ large}). \end{aligned}$$

Symbols:

- λ is the Poisson mean and variance.

Lecture 21: Bayesian Inference

Bayesian Posterior

Use: Use when an unknown parameter is modeled as random.

Formula:

$$\begin{aligned} p_{\Theta|X}(\theta | x) &= \frac{p_{\Theta}(\theta)p_{X|\Theta}(x | \theta)}{\sum_u p_{\Theta}(u)p_{X|\Theta}(x | u)}. \\ f_{\Theta|X}(\theta | x) &= \frac{f_{\Theta}(\theta)f_{X|\Theta}(x | \theta)}{\int f_{\Theta}(u)f_{X|\Theta}(x | u) du}. \end{aligned}$$

Symbols:

- Θ is the unknown random parameter.
- $X = x$ is the observed data.

MAP and LMS Estimates

Use: Use MAP for most likely posterior value and LMS for squared-error loss.

Formula:

$$\begin{aligned} \hat{\theta}_{MAP} &= \arg \max_{\theta} p_{\Theta|X}(\theta | x) \quad \text{or} \quad \arg \max_{\theta} f_{\Theta|X}(\theta | x). \\ \hat{\theta}_{LMS} &= \mathbb{E}[\Theta | X]. \end{aligned}$$

Symbols:

- MAP maximizes posterior probability/density.
- LMS minimizes mean squared error.

Lecture 22: LMS and Linear LMS Estimation

Conditional Mean as LMS

Use: Use for the best estimator under mean squared error.

Formula:

$$\begin{aligned} \hat{\Theta} &= \mathbb{E}[\Theta | X] = \arg \min_{g(X)} \mathbb{E}[(\Theta - g(X))^2]. \\ \mathbb{E}[(\Theta - \mathbb{E}[\Theta | X])^2 | X = x] &= \text{Var}(\Theta | X = x). \end{aligned}$$

Symbols:

- $g(X)$ is any estimator based on X .

LMS Error Properties

Use: Use for orthogonality and variance decomposition.

Formula:

$$\begin{aligned} \tilde{\Theta} &= \Theta - \hat{\Theta}, & \mathbb{E}[\tilde{\Theta}] &= 0, & \mathbb{E}[\tilde{\Theta}h(X)] &= 0. \\ \text{Var}(\Theta) &= \text{Var}(\hat{\Theta}) + \text{Var}(\tilde{\Theta}). \end{aligned}$$

Symbols:

- $h(X)$ is any function of the observation.

Linear LMS

Use: Use when restricting the estimator to be linear in X .

Formula:

$$\begin{aligned} \hat{\Theta}_L &= \mathbb{E}[\Theta] + \frac{\text{Cov}(\Theta, X)}{\text{Var}(X)}(X - \mathbb{E}[X]). \\ \hat{\Theta}_L &= \mathbb{E}[\Theta] + \text{Cov}(\Theta, \mathbf{X}) \text{Var}(\mathbf{X})^{-1}(\mathbf{X} - \mathbb{E}[\mathbf{X}]). \end{aligned}$$

Symbols:

- First formula is one observation.
- Second formula is vector observations.

Lecture 23: Classical Statistics

Maximum Likelihood

Use: Use when the parameter is unknown but not random.

Formula:

$$\begin{aligned} L(\theta) &= p_X(x; \theta) \quad \text{or} \quad L(\theta) = f_X(x; \theta). \\ \hat{\theta}_{ML} &= \arg \max_{\theta} L(\theta) = \arg \max_{\theta} \log L(\theta). \\ L(\theta) &= \prod_{i=1}^n p_X(x_i; \theta) \quad \text{or} \quad \prod_{i=1}^n f_X(x_i; \theta). \end{aligned}$$

Symbols:

- θ is a fixed unknown parameter.
- x_i are observed data.

Estimator Properties

Use: Use to judge bias, consistency, and mean squared error.

Formula:

$$\begin{aligned} \text{Unbiased:} \quad \mathbb{E}_{\theta}[\hat{\theta}_n] &= \theta. \\ \text{Consistent:} \quad \hat{\theta}_n &\xrightarrow{P} \theta. \\ \text{MSE}(\hat{\theta}) &= \mathbb{E}[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + \text{bias}(\hat{\theta})^2. \end{aligned}$$

Symbols:

- $\text{bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$.

Sample Mean and Confidence Interval

Use: Use for estimating an unknown mean.

Formula:

$$\begin{aligned} \bar{X} &= \frac{X_1 + \dots + X_n}{n}, & \mathbb{E}[\bar{X}] &= \mu, & \text{Var}(\bar{X}) &= \frac{\sigma^2}{n}. \\ \bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} & \text{ is a } (1 - \alpha) \text{ CI when } \sigma \text{ is known.} \end{aligned}$$

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Symbols:

- $z_{\alpha/2}$ satisfies $\mathbb{P}(Z > z_{\alpha/2}) = \alpha/2$.
- S_n^2 estimates σ^2 .

Lecture 24: Regression and Likelihood Ratio Tests

Simple Linear Regression

Use: Use to fit a line $y = \beta_0 + \beta_1 x$ by least squares.

Formula:

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{\beta_0, \beta_1} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2.$$

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Symbols:

- (x_i, y_i) are data points.
- $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$.

Likelihood Ratio Test

Use: Use for binary hypothesis testing between H_0 and H_1 .

Formula:

$$\Lambda(x) = \frac{p_X(x | H_1)}{p_X(x | H_0)} \quad \text{or} \quad \frac{f_X(x | H_1)}{f_X(x | H_0)}.$$

Choose H_1 if $\Lambda(x) > \eta$.

MAP threshold: $\eta = \frac{\mathbb{P}(H_0)}{\mathbb{P}(H_1)}.$

Symbols:

- η controls the tradeoff between error types.

Error Probabilities

Use: Use to measure false alarm and missed detection.

Formula:

$$\alpha = \mathbb{P}(\text{reject } H_0 | H_0), \quad \beta = \mathbb{P}(\text{accept } H_0 | H_1).$$

Symbols:

- α is Type I error.
- β is Type II error.

Lecture 25: Hypothesis Testing

Normal Mean Test

Use: Use LRT for normal data with different means and known variance.

Formula:

$$H_0 : X_i \sim N(\mu_0, \sigma^2), \quad H_1 : X_i \sim N(\mu_1, \sigma^2).$$

$$\log \Lambda(x) = \frac{\mu_1 - \mu_0}{\sigma^2} \sum_{i=1}^n x_i - \frac{n(\mu_1^2 - \mu_0^2)}{2\sigma^2}.$$

$$\mu_1 > \mu_0 : \quad \text{reject } H_0 \text{ if } \sum_i X_i > c.$$

Symbols:

- c is chosen from the desired significance level.

Normal Variance Test

Use: Use when testing variance of zero-mean normal data.

Formula:

$$H_0 : X_i \sim N(0, \sigma_0^2), \quad H_1 : X_i \sim N(0, \sigma_1^2).$$

$$\sum_{i=1}^n \frac{X_i^2}{\sigma_0^2} \sim \chi_n^2 \quad \text{under } H_0.$$

Symbols:

- χ_n^2 is chi-square with n degrees of freedom.

Chi-Square Goodness of Fit

Use: Use to test whether observed bin counts match expected probabilities.

Formula:

$$T = \sum_{i=1}^k \frac{(N_i - np_i)^2}{np_i}.$$

$$T \approx \chi_{k-1}^2 \quad \text{under } H_0 \text{ when } n \text{ is large.}$$

Symbols:

- N_i is observed count in bin i .
- np_i is expected count in bin i .
- k is number of bins.

Kolmogorov–Smirnov Statistic

Use: Use to compare an empirical CDF with a proposed CDF.

Formula:

$$D_n = \max_x |F_X(x) - \hat{F}_X(x)|.$$

Symbols:

- F_X is the proposed CDF.
- $\hat{F}_X$ is the empirical CDF.

Quick Symbol Reference

n	number of trials, observations, or steps
k	successes, arrivals, bins, or target count
p	Bernoulli success probability
λ	Poisson arrival rate
μ	mean, or service rate in queueing context
σ^2	variance
$N(t)$	Poisson count by time t
P	Markov transition matrix
π	steady-state distribution
θ	unknown parameter
$\hat{\theta}$	estimator of θ
α	Type I error / significance level
β	Type II error